# HARIRAMAKRRISHNAN RAMACHANDRAN

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## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Developed and deployed **Python** scripts to automate data extraction and transformation processes, reducing manual effort by approximately **40%** for the reporting team.
- Designed and implemented **Java Spring Boot** microservices to support new feature development, including building RESTful APIs for inter-service communication and data access.
- Managed production incidents for enterprise applications, performing root cause analysis and coordinating fixes to meet stringent Service Level Agreements (SLAs) across **5 critical systems**.
- Engineered and maintained **data pipelines** using SQL and Python, processing and validating approximately **15,000 daily records** from various sources for downstream analytics.
- Contributed to the research and development of **reinforcement learning algorithms** for multimodal AI, focusing on agentic behavior and decision-making in complex environments.
- Designed and implemented scalable **distributed training infrastructure** for multimodal AI systems, supporting parallel processing across multiple compute nodes.
- Containerized AI research applications using **Docker** and managed deployments on **AWS EC2** instances, enabling consistent and reproducible research environments.

## SKILLS

**AI/ML** – Reinforcement Learning, Multimodal AI, Agentic AI, Generative Models, Autoregressive Models, Diffusion Models, Policy Optimization, Reward Modeling, Exploration Strategies, Sample Efficiency, PyTorch, scikit-learn

**Programming** – Python, Java, SQL, Bash Scripting

**Backend Development** – Spring Boot, REST APIs, Microservices, FastAPI

**Cloud & DevOps** – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux

**Data Engineering** – ETL, Data Pipelines, Data Warehousing, PostgreSQL, MySQL, Oracle, pandas, NumPy

## PROJECTS

### Autonomous Agent Simulation Environment

- Developed a Python-based simulation environment to test and evaluate autonomous AI agents in dynamic scenarios, processing over **1,200 simulation steps per second**.
- Implemented novel reward mechanisms within the simulation to guide agent learning, focusing on long-term objective alignment and mitigating reward hacking.
- Engineered data collection pipelines to capture agent behavior, state transitions, and reward signals for offline analysis and model retraining.

### Multimodal Data Integration Pipeline

- Built a scalable Python pipeline to ingest and pre-process multimodal data (text, images) for AI model training, handling approximately **10 GB of data daily**.
- Integrated various data sources and applied transformation logic using pandas and NumPy, ensuring data consistency and format compatibility for downstream ML tasks.
- Designed a data validation layer to ensure the integrity and quality of curated datasets, catching common formatting errors before model ingestion.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)